Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI3301-WE01

Title: MEng and BEng Engineering

Structures and Geomatics 3

Time Allowed:	2 hours	
Additional Material provided:	None	
Materials Permitted:	(Open Note Examination).	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.	
	Revision:	

Question 1

Consider the beam in Figure Q1.1 with a pin support at point *B* and a roller support at point *D* (that cannot lift off). Assume that Young's modulus (*E*) and moment of inertia (*I*) are constant throughout the length of the beam.

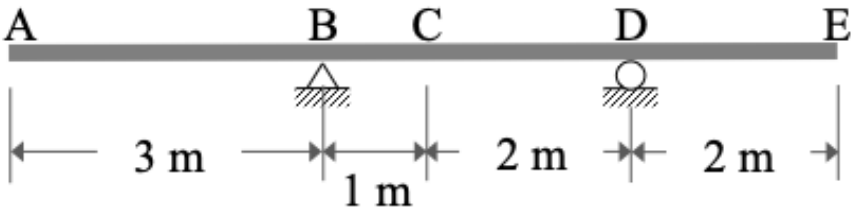


Figure Q1.1

(a) Draw the influence line for the internal moment at point *C*, which is *1 m* to the right of the support at point *B*. (the final answer should be a quantitative influence line, but any method can be used to determine it).

[25%]

(b) Based on the influence line created for *Part (a)*, explain where a continuous distributed load of *3 kN/m* would be applied (specify the starting and ending points) to create the maximum negative moment at point *C* and calculate the magnitude of this negative moment using the influence line.

[20%]

(c) Explain the difference between the influence line created for *Part (a)* and a bending moment diagram for the same beam, both in terms of what it physically represents and how it would be used in practice.

[10%]

(d) Now assume that the beam in Figure Q1.1 had the pin at point *B* replaced by a roller support. How does this change affect the analysis of the beam?

[10%]

continued

Now, consider the load case of a *1 kN* concentrated force applied at point *E* at a *45 degree* angle, as shown in Figure Q1.2. Assume that cross-sectional area (*A*), Young's modulus (*E*), and moment of inertia (*I*) are all constant throughout the length of the beam.

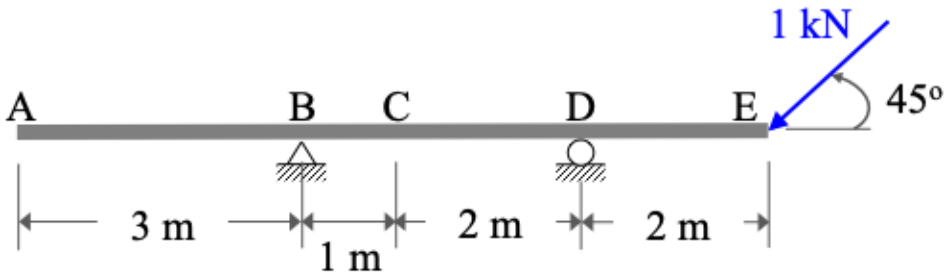


Figure Q1.2

(e) If the Geometric Stiffness Approach were to be used to analyse the beam:

i. Show a suitable discretization (nodes, elements, and degrees of freedom) to analyse the beam.

ii. Show all element component matrices needed to construct the geometric stiffness matrices for the eigenvalue problem (do not perform any of the matrix multiplication or assembly).

iii. Explain all steps needed to complete the analysis to estimate the critical loading and mode (i.e., shape) of failure due to buckling.

[35%]

Question 2

Consider the continuous beam in Figure Q2.1 with a fixed support at the right end and an intermediate pin support and loading as shown. Assume that the material and section properties are constant throughout the length of the beam with a plastic moment of $M_p = 150\ kNm$ and neglect axial loads and deformation.

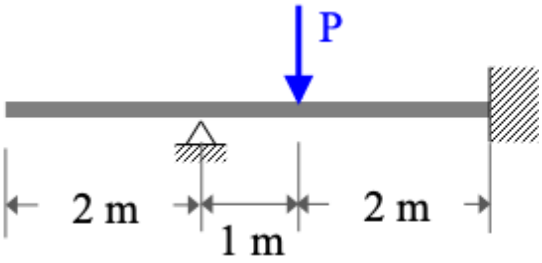


Figure Q2.1

(a) Sketch a qualitative elastic bending moment diagram (i.e., there is no need to label values), and based on this diagram, explain the expected mode of collapse.

[10%]

(b) Calculate the value of the load *P* that would lead to plastic collapse of the beam.

[30%]

(c) Draw the quantitative (i.e., labeling all minima and maxima) shear force and bending moment diagrams for the beam at the point of collapse.

[30%]

Now, consider the continuous slab in Figure Q2.2 with a pin support on the left side, a roller support on the right side, and two intermediate roller supports as shown. Assume that a uniform distributed load is applied continuously over the entire top surface, the slab experiences one-way bending, and the properties of the slab are constant and isotropic throughout.

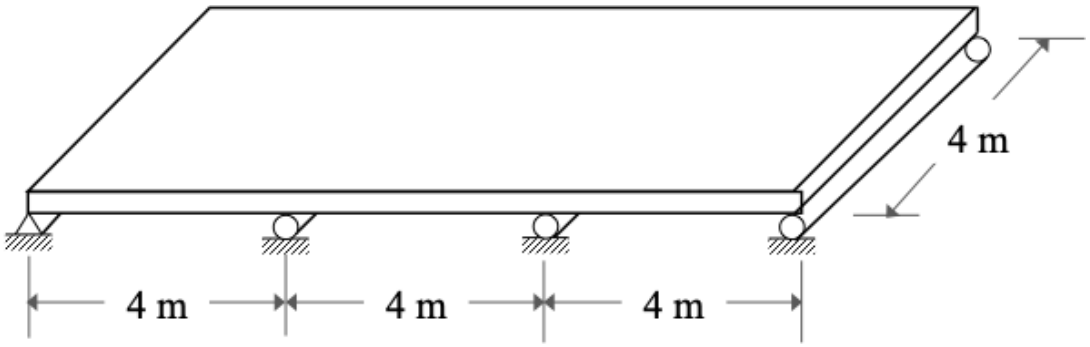


Figure Q2.2

(d) Sketch all possible collapse mechanisms and explain the procedure to determine the critical loading based on these collapse mechanisms.

[20%]

(e) Explain how the expected collapse mechanism would change if the left side was rigidly supported instead of being pin-supported.

[10%]

continued

Question 3

a) Traversing

Table Q3.1 gives the corrected bearings and uncorrected lengths of a closed loop traverse. Determine the latitude and departure error, apply the necessary adjustments and determine the coordinates of stations B-E assuming that station A has coordinates 1000,1000m.

Table Q3.1. Traverse data

Line	reduced bearing	length (m)
AB	N 20° 31' 30" E	632
BC	N 2° 44' 00" W	1031
CD	S 59° 56' 00" E	378
DE	S 08° 28' 30" W	884
EA	S 33° 31' 00" W	669

[60%]

b) Triangulation

A braced quadrilateral triangulation framework is presented in Figure Q3. The corrected internal angle measurements for the framework are presented in Table Q3.2. The horizontal distance from station B to station C has been measured as 494.8m. Determine the length of lines AB and DC.

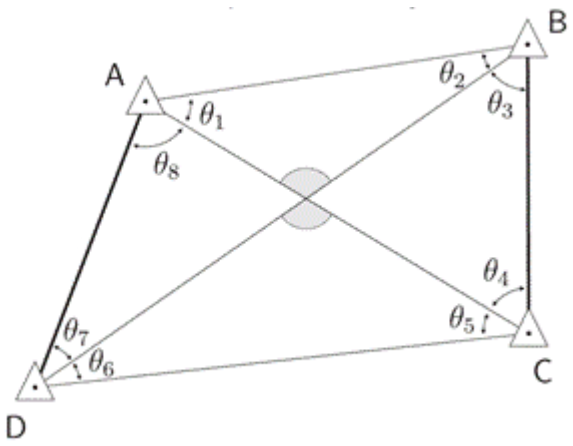


Figure Q3. Triangulation framework (not to scale).

continued

Table Q3.2. Triangulation data.

Angle	Corrected value
Θ1	22˚12’07”
Θ2	44˚40’58”
Θ3	67˚43’57”
Θ4	45˚22’58”
Θ5	37˚14’19”
Θ6	29˚38’48”
Θ7	63˚19’35”
Θ8	49˚47’30”

[10%]

c) Errors in Surveying

Describe the three principal sources of error in civil engineering surveying and give two examples of each type of these errors as might be experienced when taking tacheometry readings.

[30%]

END